Chapter One

Introduction to Emerging Technology

Topics Covered

- Emerging Technology Concepts
  - ✓ Definition
  - ✓ Currently Available Emerging Technologies
- Industrial Revolutions(IR)
  - ✓ Important Inventions of IR
  - ✓ Types of IR (the 4 IRs)
- Role of data for Emerging Technologies
- Enabling devices and networks for Emerging Technologies
- Human to Computer Interaction

1.1 Emerging Technology Concepts

- **Technology?**
  - ✓ The sum of techniques, skills, methods and process for
    - the production of goods and services
    - the accomplishment of objectives such as scientific investigations.
    - application of scientific knowledge for practical purposes
  - ✓ Branch of knowledge dealing with engineering or applied sciences.
- **Emerging Technology?**
  - ✓ Emerging technology is a term generally used to describe **a new technology**
  - ✓ But it may also refer to the *continuing development of existing technology*
  - ✓ Commonly refers to technologies that are **currently developing**, or that are **expected to be available** within the next five to ten years.

Cont. . .

- In general Emerging Technology refers to the ***technological revolution*** which is a theory of radical transformation of society through technological development.

- **Revolution?**
  - ✓ A major, sudden, and hence typically violent alteration in government and in related associations and structures. *{Social & Political Science}*
  - ✓ A radical and profound change in economic relationships and technological conditions. *{Industrial / Technology Revolution}*
- *Design for disruption :* What is expected from Engineers of today?

Cont. . .

List of Emerging Technology trends for 2020/21

- ✓ Artificial Intelligence and Machine Learning
- ✓ Robotic Processor Automation (RPA)
- ✓ Internet of Things (IoT)
- ✓ Cloud Computing and Big Data
- ✓ Enhanced Edge Computing
- ✓ Quantum Computing
- ✓ Augmented Reality and Virtual Reality
- ✓ Blockchain and Cyber Security
- ✓ Angular and React
- ✓ DevOps
- ✓ Intelligent Apps (I-Apps)
- ✓ 5G Networks
- ✓ Autonomous Devices
- ✓ Augmented Analytics
- ✓ Digital Twins
- ✓ Immersive Experiences in Smart Spaces

1.2 Industrial Revolution (IR)

- Begin during the late 1700s and early 1800s
  - ✓ The revolution started in England (increased food production, mechanized textile production)
  - ✓ The first European countries to be industrialized after England were Belgium, France, and the German state
- The IR was a time when the **manufacturing of goods** moved from:
  - ✓ small shops and homes to large factories.
- This shift brought about changes in ***culture*** as people moved from rural areas to big cities in order to work.
- IR is described as a transition to new manufacturing processes.

Cont.... .

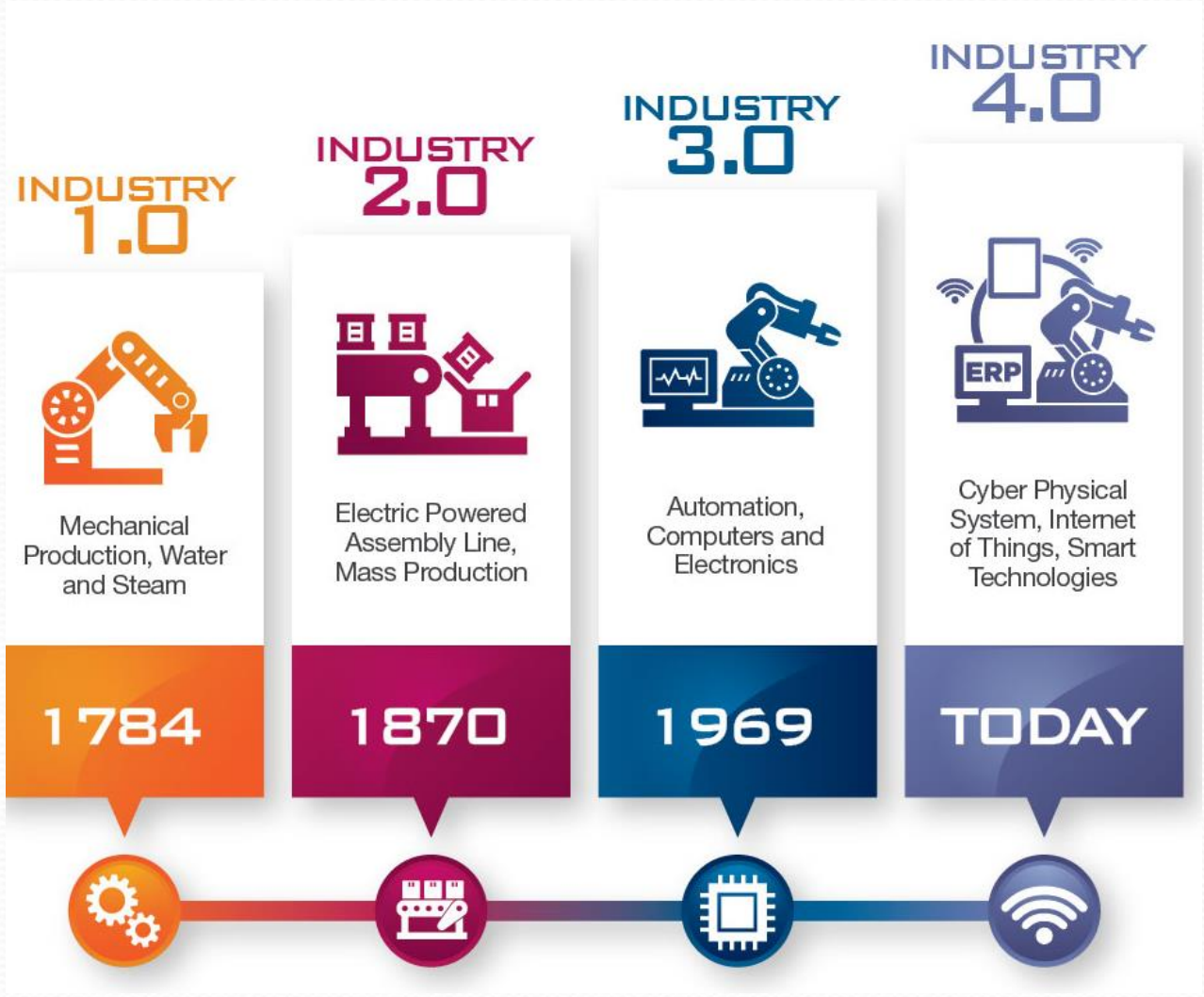
The most common Inventions of the IR

- **Transportation:** *The Steam Engine, The Railroad, The Diesel Engine, The Airplane.*

- **Communication:** *Telegraph, Transatlantic Cable, Phonograph, Telephone.*

- **Industry:** *Cotton Gin, Sewing Machine, Electric Lights.*

The 4 IRs'



Early life style of Human Being

- Hunting, gathering fruits and vegetables.
- Does not settle down at one place.
- Then to agrarian society.

IR 1.0 (Agriculture to industry: end of 18th C to 1870)

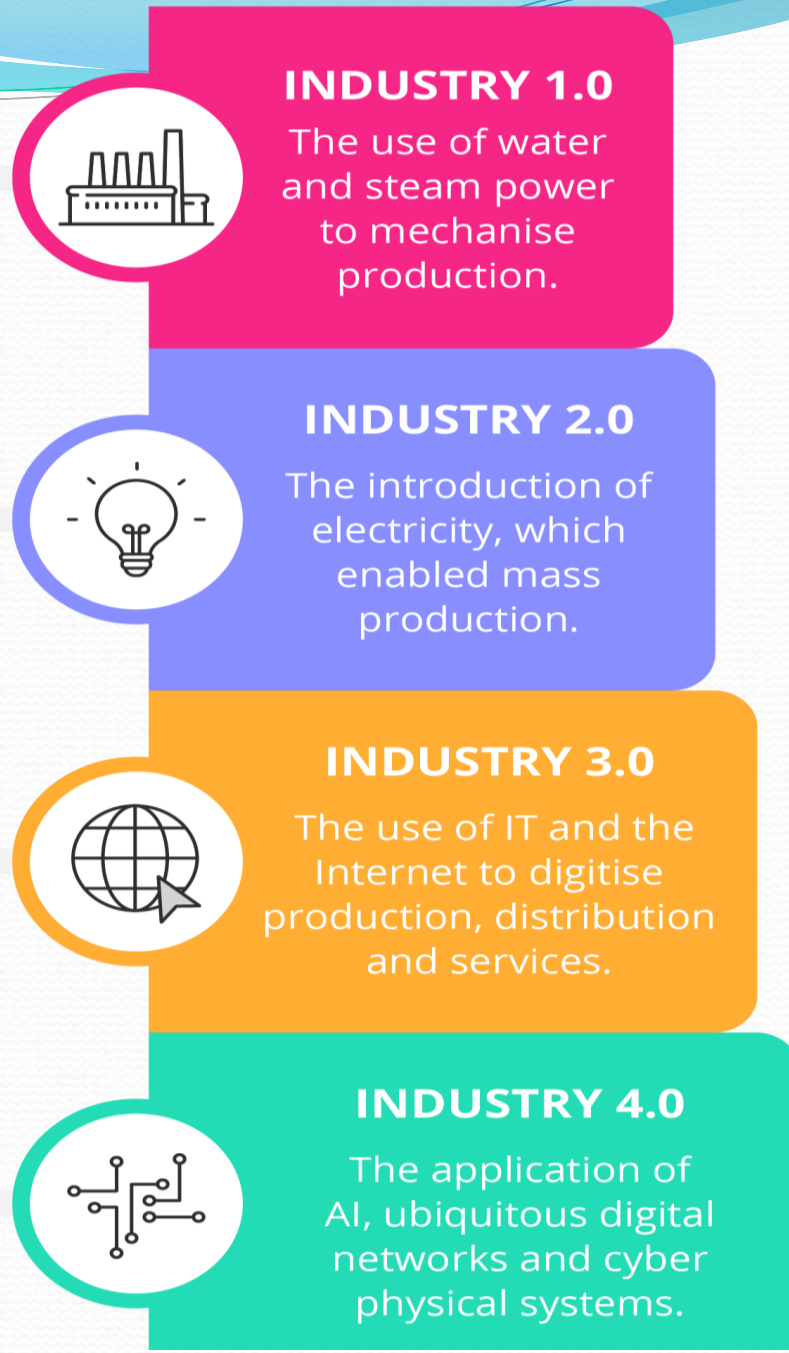
- Going from hand production methods to machines, the increasing use of steam power
- Transition from a rural based agrarian society to urban based industrial society.

The 4 IRs'

IR 2.0 (Technological/industry Revolution)

- Development of methods for manufacturing interchangeable parts
- Adoption of pre-existing technological systems such as telegraph and railroad networks.
- Vast movement of people and ideas, enhancing communication.
- New technological systems
  - ✓ **electrical power**
  - ✓ telephones
  - ✓ seed drill and new fertilizers

✓ Technological competition during and post WW1, WW2, and Cold war.



The 4 IRs'

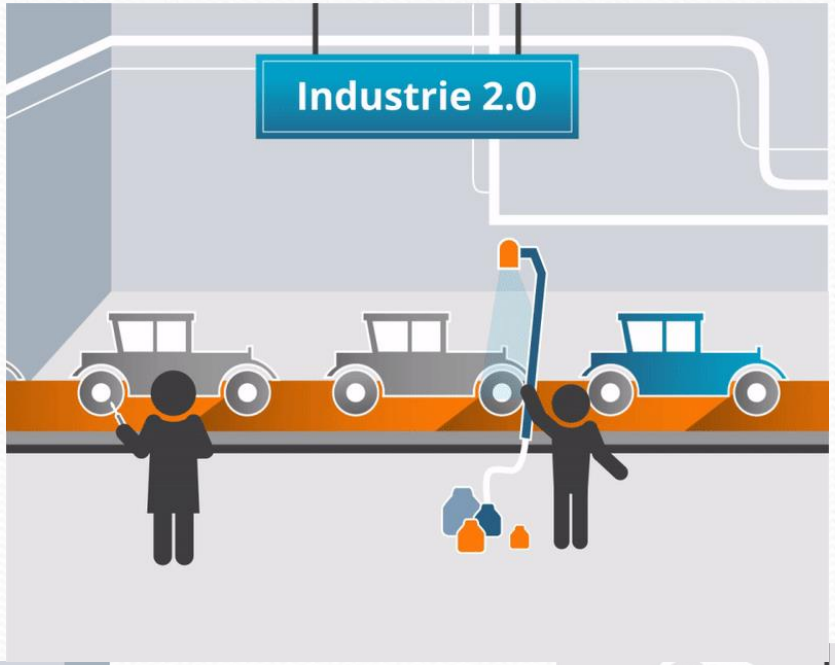
IR 3.0 (Digital/information Revolution)

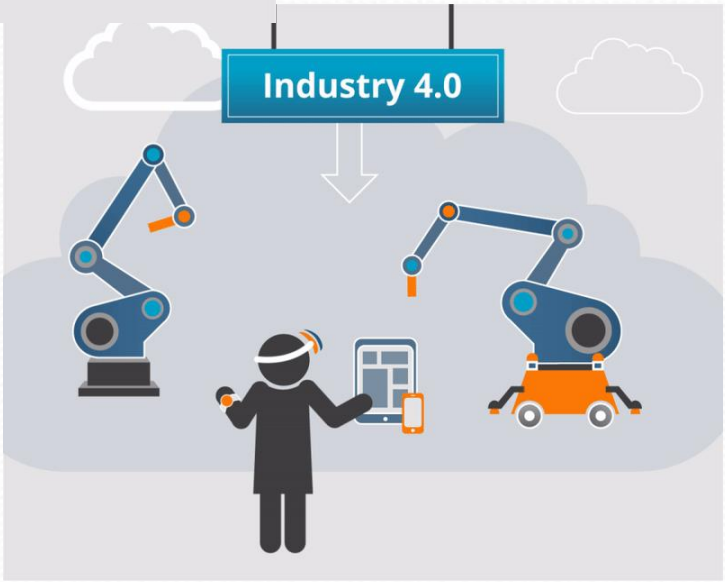
- Shift towards digitalization
- The transition from mechanical and analog electronic technology to digital electronics
- Mass production and use of digital logic circuits
- Technologies such as
  - ✓ Computer system, hand-phones and the Internet
- Digital revolution was driven by three factors:
  - ✓ information-based occupations, Cheap personal computers , **the Internet**

IR 4.0 (knowledge/Smart revolution)

- Advancements in various technologies such as *Artificial Intelligence (AI), robotics, Internet of Things (IoT), Big data*
- Additive manufacturing and autonomous vehicles
- Cyber-physical system - a mechanism that is controlled or monitored by computer-based algorithms

Summary of the 4 IRs’

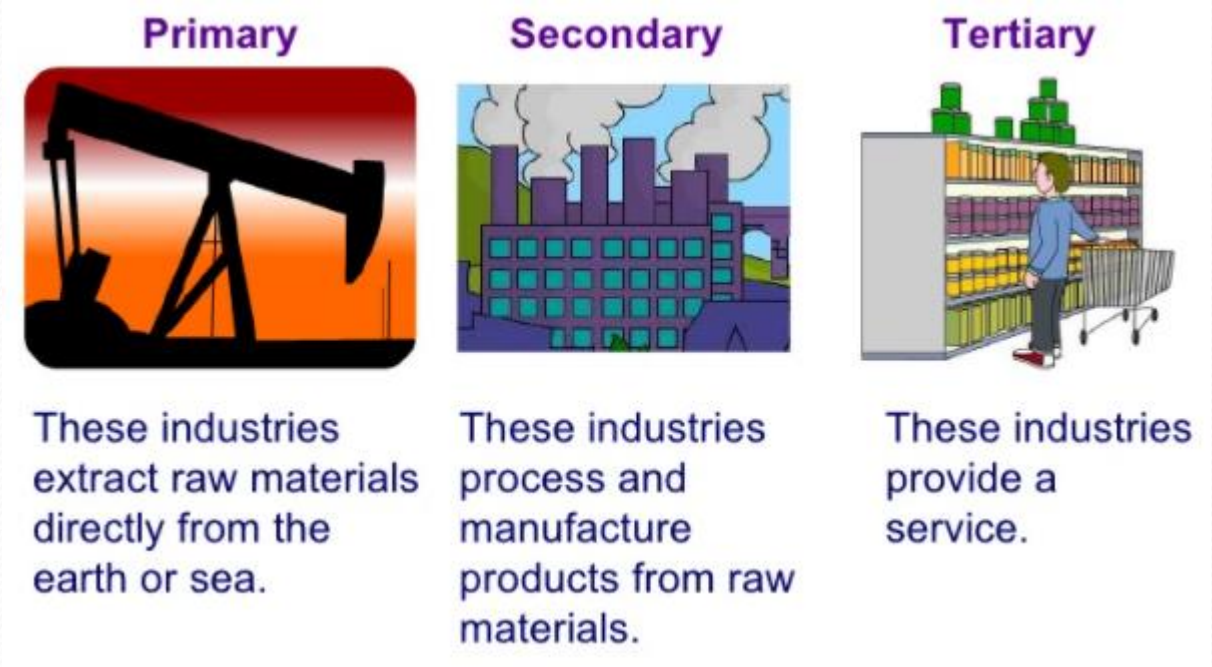


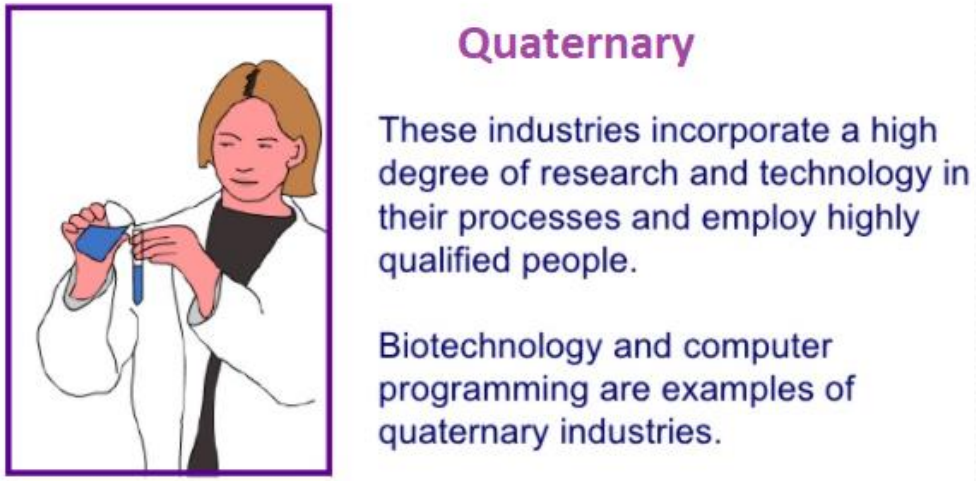


Types of industries:

- **There are four types of industries:**

1. **Primary industry:**
   - ✓ involves getting raw material
   - ✓ e.g. mining, farming, and fishing.
2. **Secondary industry:**
   - ✓ involves manufacturing
   - ✓ e.g. making cars and steel.
3. **Tertiary industries:**
   - ✓ provide a service
   - ✓ e.g. teaching and nursing.
4. **Quaternary industry:**
   - ✓ Involves research and development industries
   - ✓ e.g. IT





1.3 Role of data for Emerging Technologies

- Data is regarded as the new oil and strategic asset
- Data drives or even determines the future of science, technology, the economy, and possibly everything in our world today and tomorrow.
- The reshaping and paradigm shifting is driven not just by data itself but all other aspects that could be created, transformed, and/or adjusted by understanding, exploring, and utilizing data.
- Emerging technologies – result of data intensive scientific discovery
  - ✓ Fourth industrial revolution.
- The potential of data science and analytics to enable data-driven theory, economy, and professional development is increasingly being recognized.

1.4 Enabling device and networks for Emerging Technologies

- The basic digital devices that enables emerging technology includes

1. Memory devices – used store random information as a spreadsheet or database.
2. Microprocessors - execute software instructions to perform a wide variety of tasks
   - ✓ E.g. running a word processing program or video game
3. Logic devices - provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations etc.
   - ✓ Logic devices can be classified into two broad categories - fixed and programmable
   - ✓ **Fixed logic devices –** contain fixed circuits and permanent,
     - Perform one function or set of functions
     - Once manufactured, they cannot be changed.
   - ✓ **Programmable logic devices (PLD) –** used to build reconfigurable digital circuits
     - not limited to specific functionality
     - must be programmed before used, so that it reconfigured by using a specialized program

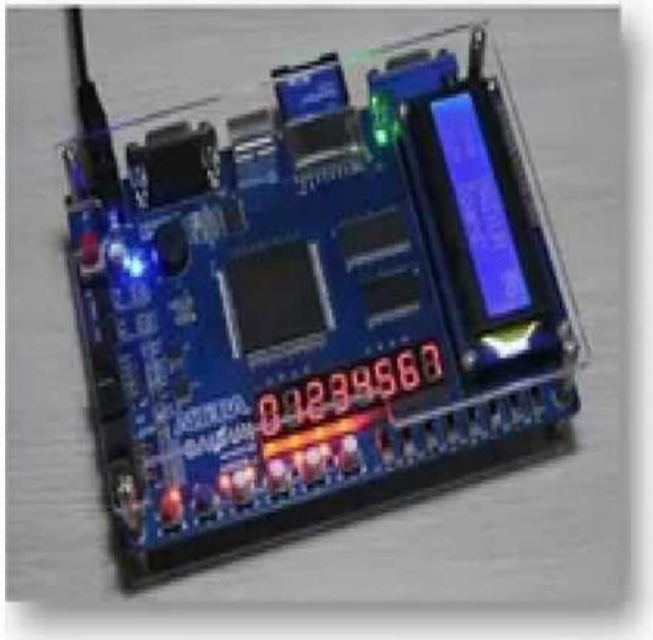
Cont…..

- PLDs are standard, off-the-shelf parts
  - ✓ Offer customers a wide range of logic capacity, features, speed, and voltage characteristics
  - ✓ Can be changed at any time to perform any number of functions.
- Simple programmable logic devices (SPLD) : PALs, PLAs, GALs, ROMs
  - ✓ The simplest, smallest and least-expensive forms of programmable logic devices
  - ✓ Use either fuses or non-volatile memory cells (EPROM, EEPROM, FLASH, and others) to define the functionality.
- Field programmable gate array(FPGA) : integrated circuit designed to be configured by a customer or a designer after manufacturing – hence the term field-programmable.
  - ✓ hardware description language (HDL)
- Complex programmable logic devices (CPLD): a programmable logic device with complexity between that of PALs and FPGAs

Cont…..

- Some uses of the programmable logic devices
  - ✓ Allow designers use inexpensive software tools to quickly develop, simulate, and test I
  - ✓ Device-to-device interfacing such as *Data communication, Signal processing, Data display, Timing, Control operations*

Altera MAX 7000-series CPLD



Different FPGAs

Computer Networks

- Interconnectivity of digital devices
- key contributing factors of the Information Technology
- History of Computer Networks
  - ✓ 1957 – Russia launched their SPUTNIK Satellite in Space
  - ✓ The American started an agency named Advance Research Project Agency (ARPA) and launched their 1st satellite within 18 month after establishment.
  - ✓ Then sharing of the information in another computer they use ARPANET.
  - ✓ Then in 1969, ARPANET comes in INDIA and INDIAN switched this name to NETWORK.
  - ✓ Development of the network began in 1969, based on designs developed during the 1960s. The ARPANET evolved into the modern Internet.
  - ✓ In the 1980s, the growth of client/server LAN architectures continued while that of mainframe computing environments declined.

Cont. . .

- **Future Trend of Networks** (Networking technology: Keep up with the changes)
  - ✓ 5G technology
  - ✓ Network developments in edge computing
  - ✓ Rise of decentralization
  - ✓ Changing perspectives on ML and AI
  - ✓ More attention to network security
  - ✓ Going wireless
  - ✓ Cloud repatriation
  - ✓ Smart automation expands

1.5 Human to Machine Interaction (HMI)

- **HMI** is a discipline concerned with the design, evaluation an implementation of interactive computing systems for human use and with the study of major phenomena surrounding them. *(Association for Computing Machinery (ACM))*

- Shortly HMI refers to the communication and interaction between a human and a machine via a user interface.
  - ✓ Via Hardware for: *Input and output – display*
  - ✓ Sometimes termed human–computer interaction (HCI), man-machine interaction (MMI) or computer-human interaction (CHI).

- HCI – is the study of how people interact with computers and to what extent could computers interact with human beings successfully.
- HCI improves the interaction between users and computers by making computers more user-friendly and receptive to the user's needs.

Cont. . .

- An important facet of HCI is user satisfaction (or simply End User Computing Satisfaction).
- It draw from supporting knowledge on both the machine and the human side.
- The main advantages of HCI are
  - simplicity,
  - ease of deployment & operations and
  - cost savings for smaller set-ups,
  - reduce solution and design time and
  - reduce integration complexity

Cont….

- The most common means of interacting with computer includes
  - ✓ Graphical User Interface (GUI)
  - ✓ Command Line Interface (CLI)

- HCI consists of three parts:
  - ✓ The user,
  - ✓ The computer/machine itself, and
  - ✓ The ways they work together (Know-how)

Cont…

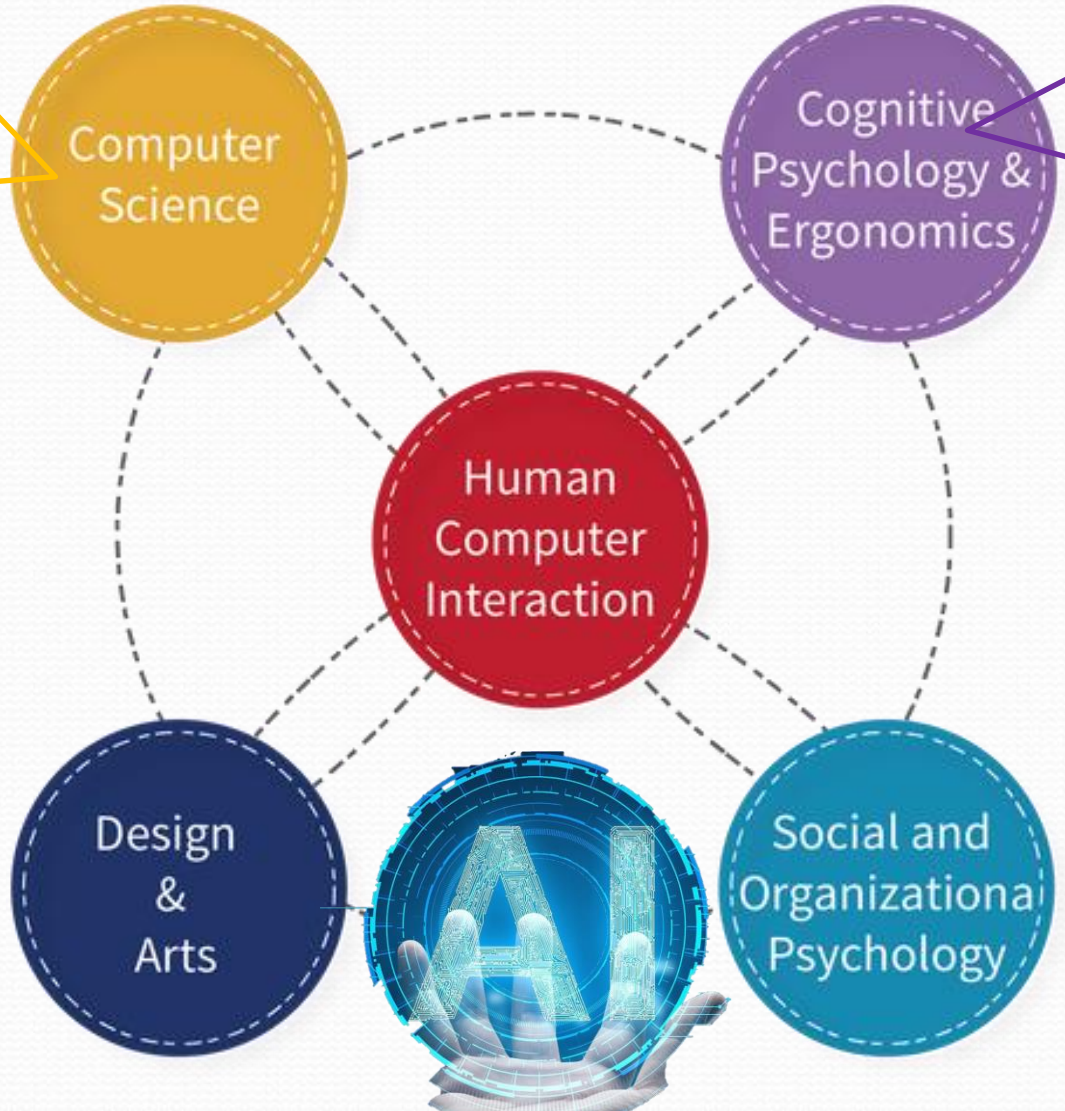
- Poorly designed human-machine interfaces can lead to many unexpected problems.
- Classical examples
  - ✓ Three Mile Island accident in USA,
  - ✓ a nuclear meltdown accident

✓ Therefore the HCI must be

Cont….

The most common multidisciplinary fields of HCI

- graphics,
- technology,
- prototyping tools,
- user interface management systems



- Limitations,
- information processing,
- performance prediction,
- cooperative working, and capabilities.

Cont….

- The **concepts of human-computer interaction** include the following
  - ✓ User customization - users are the important sources of new applications
  - ✓ Embedded computation - little computations and automated processes
  - ✓ Augmented reality - notion of layering relevant information into our vision of the world
  - ✓ Social computing - individuals perceive their interactions with computers more positively than humans, despite behaving the same way towards these machines.
  - ✓ Knowledge-driven human–computer interaction - semantic gap usually exists between human and computer's understandings towards mutual behaviors.
  - ✓ Emotions and human-computer interaction - affect-detection channels have the potential to detect human emotions
  - ✓ Brain–computer interfaces - direct communication pathway between an enhanced or wired brain and an external device.

End of Chapter 1
(Int. to Emerging Technology)

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Summary Questions

1. What is Emerging Technology?
2. Given an example of currently emerged and future trends of emerging technologies.
3. Mention the most important inventions of industrial revolutions?
4. List and discuss the 4 Industrial revolutions.
5. What is the Role of Data for Emerging Technologies?
6. List out programmable devices and discuss their features
7. What is HCI (Human-Computer Interaction)?
8. Write disciplines that contribute to Human-Computer Interaction

Thank you!